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- To compile the client.c file use with command "gcc client.c -o client"
- To compile the server.c file use the command "gcc server.c -o server"
- To run the client file do ./client IP:PORT output_file_name (which in our case will be mydata_client_copy.txt)
 - IP and PORT are the server's IP and the PORT server is listening on.
- To run the server file do ./server PORT input_file_name

Showing the demo below

- IP of vm1 where client.c is located, 10.10.1.2

```
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.1.2 netmask 255.255.255.0 broadcast 10.10.1.255
    inet6 fe80::4e:97ff:fe74:5d79 prefixlen 64 scopeid 0x20<link>
    ether 02:4e:97:74:5d:79 txqueuelen 1000 (Ethernet)
    RX packets 28 bytes 1760 (1.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1285 (1.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- IP of vm0 where server.c is located, 10.10.1.1

```
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.1.1 netmask 255.255.255.0 broadcast 10.10.1.255
    inet6 fe80::e:19ff:fe94:d498 prefixlen 64 scopeid 0x20<link>
    ether 02:0e:19:94:d4:98 txqueuelen 1000 (Ethernet)
    RX packets 42 bytes 3069 (3.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1285 (1.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- Compiling server.c file and running it

```
abhag017@vm0:~$ vim server.c
abhag017@vm0:~$ gcc server.c -o server
abhag017@vm0:~$ vim try_test.txt
abhag017@vm0:~$ vim try_test2.txt
abhag017@vm0:~$ du -sh try*
8.0K    try_test2.txt
4.0K    try_test.txt
abhag017@vm0:~$ ./server 8000 try_test.txt
```

- ```
abhag017@vm1:~$ vim client.c
abhag017@vm1:~$ gcc client.c -o client
abhag017@vm1:~$./client 10.10.1.1:8000 mydata_client_copy.txt
```

- ```
After parsing IP, PORT and filename are 10.10.1.1, 8000, mydata_client_copy.txt
connected to 10.10.1.1:8000
Bytes received 58
Output file size is 58 bytes
abthag017@vml1:~$ shasum -a 256 mydata_client_copy.txt
4dbf380b93c13f02abc15246883065643a0f03a9493b063b3568f38121a7d144  mydata_client_copy.txt
```

- ```
Will Transfer try_test.txt to incoming connections
Bind Success to 8000
Listen Successfully on 8000
Bytes read 58
Total Transfer Done: 58 bytes
End of file, closing this connection now
^C
abha017@vm0:~$ shasum -a 256 try_test.txt
4dbf380b93c13f02abc15246883065643a0f03a9493b063b3568f38121a7d144 try_test.txt
```

- [illegible]

```

Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 256
Bytes read 207
Total Transfer Done: 8143 bytes
End of file, closing this connection now
^C
abthag017@vm0:~$ shasum -a 256 try_test.txt
4dbf380b93c13f02abc15246883065643a0f03a9493b063b3568f38121a7d144 try_test.txt
abthag017@vm0:~$ shasum -a 256 try_test2.txt
5fd6410bac71f505d0ccc679df42582dd7c2cae5c95ea7f711987f06b8384b11 try_test2.txt

```

- Same process for client.c with the big transfer file, showing output and checksum in 2 images

```

abthag017@vm1:~$./client 10.10.1.1:8000 mydata_client_copy.txt
After parsing IP, PORT and filename are 10.10.1.1, 8000, mydata_client_copy.txt
connected to 10.10.1.1:8000
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
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Bytes received 256
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Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 256
Bytes received 207
Output file size is 8143 bytes
abthag017@vm1:~$ shasum -a 256 mydata_client_copy.txt
5fd6410bac71f505d0ccc679df42582dd7c2cae5c95ea7f711987f06b8384b11 mydata_client_copy.txt

```